2 **(a)** It can be seen from:

the straight line graph passing through the origin that the acceleration a of the plate is directly proportional to the displacement x from its equilibrium position; and

the negative gradient that its acceleration a is opposite in direction to its displacement x, and towards the equilibrium position.

These two features define simple harmonic motion.

(b) **(i)** The acceleration of the plate at which the sand first loses contact with the plate is the acceleration of free fall ($-9.81\ \mathrm{m\,s^{-2}}$).

The point where the sand first loses contact with the plate is at the maximum displacement of the plate and when the plate is moving in the downward direction.

If at this moment, the plate has the same downward acceleration as that of free fall, there is no contact force between the sand and the plate.

Note: The resultant force acting on the sand is not zero when the sand lost contact with the plate.

(ii) 3.8 mm.